Who Validates the Validators? Evidence-Gated Measurement Audits for AI Benchmarks

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Abstract

Benchmark audits are often interpreted as if their diagnostics were already validated measurement instruments. We study a stricter alternative: each diagnostic must expose its sensitivity, uncertainty, practical materiality, external validity, and transfer limits before it can license a benchmark-quality or model-ranking claim. We implement this discipline in ValidEval through versioned evidence states, exact-identity contracts, leakage guards, null-calibrated ranking analyses, and fail-closed import and reporting paths. As a historical case study, we independently reconstruct a public HELM-derived MMLU response panel with 39 models, 14,042 items, and no missing cells. The reconstruction verifies descriptive rank variation but does not treat a legacy raw rank threshold as evidence of material instability. Controlled cross-benchmark, human, external-label, and decoupled-synthetic results remain explicit result requirements. The intended contribution is therefore a falsifiable measurement-audit methodology, not a scalar benchmark-validity score.

1 Introduction

A benchmark score can be perfectly computed and still fail to support the interpretation attached to it. Construct underrepresentation, shortcuts, contamination, scoring artifacts, dependent items, saturation, and weak discrimination affect different inferences and should not be collapsed into one health score. This distinction is familiar in validity theory [Messick, 1995, American Educational Research Association et al., 2014] but remains unevenly enforced in contemporary model evaluation.

ValidEval asks a prior question: who validates the diagnostics used to audit a benchmark? A warning that correlates with model disagreement is not automatically an error detector. A response matrix with many models is not automatically a valid IRT sample. A large rank range is not automatically a material decision change. Our central claim is methodological: diagnostic outputs should be linked to explicit estimands, uncertainty, validation evidence, and blocked wording.

We separate a historical imported study (Study H) from a controlled exact-checkpoint study (Study C). The current paper scaffold reports only artifact-backed Study H results. Every unexecuted study retains an exact result requirement rather than a plausible number.

2 Related Work

HELM and lm-evaluation-harness emphasize reproducible model evaluation [Liang et al., 2022, Gao et al., 2023]. BetterBench evaluates benchmark-development practices [Reuel et al., 2024], while BenchBench examines how methodological choices affect cross-benchmark agreement [Perlitz et al., 2024]. Recent work directly applies IRT to benchmark-item auditing [Land and Bikel, 2026], and tinyBenchmarks uses psychometric ideas for efficient performance estimation [Polo et al., 2024].

Table 1: V5 claim states. Counts are categorical and are not combined into a score.

State	Claims
BLOCKED	3
CONTRADICTED	1
REPRODUCED	5
RETIRED	2

Table 2: Independently reproduced Study H panel integrity.

Quantity	Reproduced value
Models	39
Items	14042
Subjects	57
Response rows	547638
Missing cells	0
Contradictory duplicates	0
Observed accuracy spread	0.580188

These works make “first IRT benchmark audit” and “first benchmark checklist” unavailable novelty claims.

Our narrower target is the validation of diagnostic instruments and the claims they license. It combines uncertainty-aware evaluation [Ye et al., 2024], exact protocol identity, and executable evidence gates. Multiple-choice option order can materially alter model outputs [Pezeshkpour and Hruschka, 2024]; this motivates treating prompts and extraction rules as part of the measured condition. Model-pretraining contamination requires separate methods and cannot be ruled out merely by securing the project pipeline [Oren et al., 2024].

3 Evidence-Gated Measurement Audits

Every claim maps to one categorical evidence state. These states distinguish reproduced primary evidence, reported but unreproduced results, fixtures, plans, blocks, contradictions, staleness, and retirement; they are deliberately not an ordinal score. A dependent analysis stops when its input identity, integrity, or sample-adequacy gate fails.

The framework distinguishes four threat classes: project-pipeline leakage, benchmark split or item overlap, diagnostic-label leakage, and uncertain model-pretraining contamination. Gold answers are not available to generation, retry, or extraction decisions. Human reviewers are blinded to diagnostic selection and external labels. Synthetic labels remain inaccessible to detector feature construction.

4 Historical Public MMLU Study

Starting from the earliest defensible local HELM-derived prediction artifact, we reconstruct a normalized long table and wide response matrix. The reconstructed matrix is byte-identical to the active cached matrix. No duplicate, contradictory, invalid-correctness, partial-coverage, or missing cell was found.

Table 3: Controlled Study C cross-benchmark gate. The table remains blocked until exact-checkpoint MMLU, GSM8K, and BBH artifacts pass V5 import and overlap checks.

Field	Artifact-derived value
Overlap gate	BLOCKED
Exact common checkpoints	[RESULT REQUIRED]
Independent families	[RESULT REQUIRED]
Minimum extraction reliability	[RESULT REQUIRED]
Transfer conclusion	BLOCKED

Raw subject-conditioned ranks have median range 19 and maximum range 30 across the 39-model panel. These values are descriptive. The earlier threshold that called range at least 10 "severe" was post hoc and is retired. V5 evaluates the observed values against explicit null models and reports confidence sets, outranking probabilities, and top-k stability. The proxy diagnostic-weighted ranking remains close to aggregate accuracy, but it is not a full 2PL estimate or a true ranking.

5 Controlled Exact-Checkpoint Study

Study C fixes the same exact model revisions, prompt policies, generation settings, extraction versions, and scoring versions across MMLU, GSM8K, and BBH. It is separate from Study H; historical family names do not create exact checkpoint overlap.

[RESULT REQUIRED: results/cross_benchmark/controlled_study_v5.json]

The preregistered estimands are exact-model rank transfer, model-by-benchmark interaction, diagnostic-family transfer, and decision stability. The analysis may conclude supported, partial, benchmark-specific, unsupported, underpowered, or blocked transfer. A universal transfer claim is not an allowed default.

Permitted populated conclusions are `TRANSFER_SUPPORTED`, `TRANSFER_PARTIAL`, `TRANSFER_BENCHMARK_SPECIFIC`, `TRANSFER_NOT_SUPPORTED`, `UNDERPOWERED`, or `BLOCKED`. Family-level analysis, when used because exact checkpoint overlap is insufficient, must be labeled exploratory and must not populate an exact-model transfer claim.

6 Validation of the Audit Instruments

Three independent validation pillars are predeclared. First, a decoupled synthetic protocol freezes generators, seeds, diagnostics, metrics, controls, and held-out flaw families before execution. Second, a blinded human protocol estimates enrichment or precision according to its sampling design; it does not infer recall from a high-score-only queue. Third, external item validation requires confirmed identity rather than subject/index alignment.

[RESULT REQUIRED: results/synthetic/confirmatory_v5/results.json]
[RESULT REQUIRED: results/human/human_validation_v5.json]
[RESULT REQUIRED: results/mmlu_redux/confirmed_validation_v5.json]

The current MMLU-Redux path lacks a direct ID or content-hash confirmation in the sanitized local artifacts. It is therefore treated as an unsuccessful exploratory linkage attempt, not validation of item-error detection.

7 Limitations, Ethics, and Responsible Evaluation

The historical panel is not a controlled sample of model systems and contains family dependence. Public-benchmark contamination in model pretraining cannot be excluded. The controlled panel may be limited by T4 memory, licenses, gated access, and quantization; every such condition must be recorded. Measurement models can fail identifiability or convergence gates. Benchmark and human-label findings remain protocol- and population-scoped.

Benchmark audits can unfairly damage datasets or model developers when exploratory flags are described as proven defects. ValidEval therefore preserves audit trails, reports uncertainty, redacts secret-like values, blinds human review, avoids paid APIs as required dependencies, and blocks global "valid/invalid" labels. Raw benchmark text and expired signed URLs found in excluded source data are not included in the public release.

8 Conclusion

Validity is not accuracy, and an audit is not validated merely because it produces plausible warnings. ValidEval makes the chain from primary artifact to claim explicit and fail closed. The historical MMLU reconstruction establishes a trustworthy descriptive substrate; the scientific ceiling now depends on controlled exact-checkpoint, human/external, and decoupled-synthetic evidence.

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